

Ian Gershon Teixeira

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Academic Appointments and Research Positions

University of California San Diego

Stefan E. Warschawski Visiting Assistant Professor of Mathematics 2025–present

University of Maryland, College Park

MathQuantum Fellow, NSF Research Training Group in the Mathematics of Quantum Information 2024–2025

University of Maryland / QuICS

Researcher, NSF Quantum Leap Challenge Institute for Robust Quantum Simulation 2022–2023

University of California, Santa Barbara

NSF REU Summer Researcher in Mathematics, Quantum Groups 2016

Education

University of Maryland, College Park

Ph.D. in Applied Mathematics

Dissertation: *Quantum Codes, Transversal Gates, and Representation Theory*

Advisors: Jonathan Rosenberg (Mathematics) and Michael Gullans (Physics) 2018–2025

Wesleyan University

B.A. in Mathematics, *Phi Beta Kappa* 2013–2017

Publications

- [1] E. Kubischta and **I. Teixeira**. “Nuclear-spin statistical weights from Young diagrams.” *Journal of Chemical Physics* **164**, 194120 (2026).
- [2] E. Kubischta and **I. Teixeira**. “Flexible fault-tolerant gate gadgets.” *Physical Review A* **113**, 022421 (2026).
- [3] E. Kubischta and **I. Teixeira**. “Classification of the subgroups of the two-qubit Clifford group.” *Journal of Mathematical Physics* **66**, 122203 (2025).
- [4] E. Kubischta and **I. Teixeira**. “Quantum codes and irreducible products of characters.” *Designs, Codes and Cryptography* **93**, 2919–2930 (2025).
- [5] E. Kubischta and **I. Teixeira**. “Permutation-invariant quantum codes with transversal generalized phase gates.” *IEEE Transactions on Information Theory* **71**(1), 485–498 (2025).
- [6] E. Kubischta and **I. Teixeira**. “Quantum codes from twisted unitary t -groups.” *Physical Review Letters* **133**, 030602 (2024).
- [7] E. Kubischta and **I. Teixeira**. “A family of quantum codes with exotic transversal gates.” *Physical Review Letters* **131**, 240601 (2023).

Preprints

- [P1] **I. Teixeira**. “ A_2 -frames, negacyclic glue, and extremal strongly 6-modular lattices.” arXiv:2609.27148 (2026).
- [P2] **I. Teixeira** and D. Meyer. “Deep Holes in the Clifford Hierarchy.” arXiv:2608.08403 (2026).
- [P3] E. Kubischta and **I. Teixeira**. “Time-Reversal Selection Rules for Quantum Error Correction.” arXiv:2608.06304 (2026).
- [P4] E. Kubischta and **I. Teixeira**. “Classical and Quantum MacWilliams Transforms as Spin Kinematics.” arXiv:2607.27173 (2026).
- [P5] E. Kubischta and **I. Teixeira**. “Minimal Permutation-Invariant Qudit Codes from Edge-Colorings of Complete Graphs.” arXiv:2605.22439 (2026).
- [P6] **I. Teixeira**. “Orthogonal Polynomials and the MacWilliams Transform for Permutation-Invariant Qudit Codes.” arXiv:2605.15372 (2026).
- [P7] E. Kubischta and **I. Teixeira**. “MacWilliams Identities for Intrinsic Quantum Codes.” arXiv:2604.16023 (2026).
- [P8] E. Kubischta and **I. Teixeira**. “Intrinsic Quantum Codes.” arXiv:2511.14840 (2025).

- [P9] E. Kubischta, **I. Teixeira**, and J. Maxwell Silvester. “Quantum Weight Enumerators for Real Codes with X and Z Exactly Transversal.” arXiv:2306.12526 (2023).

Selected Invited and Conference Talks

- **Deep Holes in the Clifford Hierarchy**, RIT on Geometry and Physics, University of Maryland, September 17, 2026. *Invited seminar.*
- **Deep Holes in the Clifford Hierarchy**, QuEra Computing, September 10, 2026. *Invited seminar.*
- **Intrinsic Quantum Codes**, QASAR Seminar on Quantum Error Correction and Fault Tolerance, INRIA, May 2026. Joint talk with Eric Kubischta. *Invited seminar.*
- **Icosahedral Quantum Codes from Twisted Unitary t -Groups**, MathQuantum Annual Symposium, University of Maryland, May 2025.
- **Icosahedral Quantum Codes from Twisted Unitary t -Groups**, Quantum Information Processing Conference (QIP 2025), Duke University, February 2025. Joint talk with Eric Kubischta.
- **Symmetry and the Reduction of the Error Correcting Conditions**, Fermilab Quantum Information Science / High Energy Physics Seminar, Fermi National Accelerator Laboratory, July 2023. Joint talk with Eric Kubischta. *Invited seminar.*

Teaching and Mentoring

- University of California San Diego** — Instructor of Record 2025–present
- MATH 103A, Modern Algebra I — Fall 2025; Winter 2026.
 - MATH 18, Linear Algebra — Winter 2026; Fall 2026.
 - MATH 102, Applied Linear Algebra — Spring 2026.
- University of Maryland** — Instructor of Record 2022
- Linear Algebra MATH 240; Precalculus MATH 113.
- University of Maryland** — Graduate Student Mentor 2022
- Directed Reading Program in quantum error correction.
- University of Maryland** — Teaching Assistant 2018–2021
- Linear Algebra MATH 240 (four semesters); Linear Algebra for Scientists and Engineers MATH 461 (two semesters); Calculus III MATH 241; Statistics STAT 100.

Awards and Fellowships

- MathQuantum Fellowship, University of Maryland, 2024.
- Hauptman Summer Fellowship, University of Maryland, 2024.
- Jacob K. Goldhaber Travel Grant, University of Maryland, 2024.
- International Conference Student Support Award, 2024.
- Dean’s Fellowship, University of Maryland, 2018–2020.
- Rice Prize, Wesleyan University, 2017 — awarded to the top senior in mathematics.

Professional Service

- Referee for *Physical Review Letters*, *Physical Review X*, *IEEE Transactions on Information Theory*, and *Physical Chemistry Chemical Physics*.